

Name: B SREE NIKHIL REDDY ROLL No: 2520090058

Practical week-2

Step 1 — Create Database

The screenshot shows the SQL Developer interface. The left sidebar displays the 'SCHEMAS' tree with 'pweek2' listed under 'sys'. The main editor window shows the command 'create database pweek2'. The 'Output' window at the bottom displays the execution results:

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec

Step 2 — Create Books Table (DDL + Constraints)

The screenshot shows the SQL Developer interface. The left sidebar displays the 'SCHEMAS' tree with 'pweek2' listed under 'sys'. The main editor window shows the command 'CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK (published_year < 2027));'. The 'Output' window at the bottom displays the execution results:

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY, ...	0 row(s) affected	0.047 sec

Step 3 & 4 — Insert Books and Display

Navigator: SQL File 3* SQL File 3* SQL File 4* SQL File 5* x SQLAdditions

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY, ...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_yea...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec

Navigator: SQL File 3* SQL File 3* SQL File 4* SQL File 5* x SQLAdditions

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY, ...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_yea...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Result Grid

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	NULL	NULL	NULL

Step 5, 6 & 7 — Members Table: Create, Insert, Display

The screenshot shows a SQL IDE interface with a Navigator on the left, a central SQL editor, and an Output pane at the bottom.

Navigator: Shows a tree view with 'SCHEMAS' and a search filter 'Filter objects'. The tree includes 'joins', 'klu', 'pweek1', and 'sys'. Below the tree are tabs for 'Administration' and 'Schemas', and a status bar indicating 'No object selected'.

SQL Editor: Contains the following SQL code:

```
6 title VARCHAR(100) NOT NULL,  
7 isbn VARCHAR(20) UNIQUE,  
8 published_year INT CHECK (published_year < 2027)  
9 );  
10  
11 • INSERT INTO Books (book_id, title, isbn, published_year) VALUES  
12 (1, 'The Great Gatsby', '9780743273565', 1925),  
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),  
14 (3, '1984', '9780451524935', 1949);  
15  
16 • SELECT * FROM Books;  
17  
18 • CREATE TABLE Members (  
19 member_id INT PRIMARY KEY,  
20 full_name VARCHAR(100),  
21 email VARCHAR(100) UNIQUE  
22 );  
23
```

Output: The 'Action Output' pane shows the execution results of the SQL statements:

#	Time	Action	Message	Duration / Fetch
✓ 1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
✓ 2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
✓ 3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY,...	0 row(s) affected	0.047 sec
✓ 4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_ye...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
✓ 5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	15:47:19	CREATE TABLE Members (member_id INT PRIMARY...	0 row(s) affected	0.047 sec

Navigator: SCHEMAS
Filter objects
joins
klu
pweek1
sys

Administration Schemas
Information

No object selected

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x
Limit to 1000 rows

```

12 (1, 'The Great Gatsby', '9780743273565', 1925),
13 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
14 (3, '1984', '9780451524935', 1949);
15
16 • SELECT * FROM Books;
17
18 • CREATE TABLE Members (
19     member_id INT PRIMARY KEY,
20     full_name VARCHAR(100),
21     email VARCHAR(100) UNIQUE
22 );
23
24 • INSERT INTO Members (member_id, full_name, email) VALUES
25 (101, 'John Smith', 'john.smith@email.com'),
26 (102, 'Emma Wilson', 'emma.wilson@email.com'),
27 (103, 'Michael Brown', 'michael.brown@email.com');
28
29 • SELECT * FROM Members;

```

SQLAdditions My Snippets

Context Help Snippets

Output

#	Time	Action	Message	Duration / Fetch
✓ 1	15:41:57	create database pweek2	1 row(s) affected	0.000 sec
✓ 2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
✓ 3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY,...	0 row(s) affected	0.047 sec
✓ 4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_ye...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
✓ 5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	15:47:19	CREATE TABLE Members (member_id INT PRIMARY...	0 row(s) affected	0.047 sec
✓ 7	15:47:47	INSERT INTO Members (member_id, full_name, email) ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec

Object Info Session

Navigator: SCHEMAS
Filter objects
joins
klu
pweek1
sys

Administration Schemas
Information

No object selected

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x
Limit to 1000 rows

```

20 full_name VARCHAR(100),
21 email VARCHAR(100) UNIQUE
22 );
23
24 • INSERT INTO Members (member_id, full_name, email) VALUES
25 (101, 'John Smith', 'john.smith@email.com'),
26 (102, 'Emma Wilson', 'emma.wilson@email.com'),
27 (103, 'Michael Brown', 'michael.brown@email.com');
28
29 • SELECT * FROM Members;

```

SQLAdditions My Snippets

Context Help Snippets

Result Grid

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com
* NULL	NULL	NULL

Members 2 x Apply Revert

Output

#	Time	Action	Message	Duration / Fetch
✓ 2	15:44:04	use pweek2	0 row(s) affected	0.000 sec
✓ 3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KE...	0 row(s) affected	0.047 sec
✓ 4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_y...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
✓ 5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	15:47:19	CREATE TABLE Members (member_id INT PRIMAR...	0 row(s) affected	0.047 sec
✓ 7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
✓ 8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 8 — Create Loans Table (FOREIGN KEYS)

The screenshot shows a SQL IDE interface with a Navigator on the left, a central SQL editor, and an Output window at the bottom.

Navigator: Shows a tree view with folders for 'joins', 'klu', 'pweek1', and 'sys'. The 'Schemas' tab is active.

SQL Editor: Contains the following SQL code:

```
21 email VARCHAR(100) UNIQUE
22 );
23
24 • INSERT INTO Members (member_id, full_name, email) VALUES
25 (101, 'John Smith', 'john.smith@email.com'),
26 (102, 'Emma Wilson', 'emma.wilson@email.com'),
27 (103, 'Michael Brown', 'michael.brown@email.com');
28
29 • SELECT * FROM Members;
30
31 • CREATE TABLE Loans (
32 loan_id INT PRIMARY KEY,
33 member_id INT,
34 book_id INT,
35 loan_date DATE,
36 FOREIGN KEY (member_id) REFERENCES Members(member_id),
37 FOREIGN KEY (book_id) REFERENCES Books(book_id)
38 );
```

Output Window: Displays the execution results of the SQL statements.

#	Time	Action	Message	Duration / Fetch
3	15:44:11	CREATE TABLE Books (book_id INT PRIMARY KEY...	0 row(s) affected	0.047 sec
4	15:45:08	INSERT INTO Books (book_id, title, isbn, published_y...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	15:45:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	15:47:19	CREATE TABLE Members (member_id INT PRIMAR...	0 row(s) affected	0.047 sec
7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec

Step 9 & 10 — Insert 10 Loans and Display

Navigator: SCHEMAS
Filter objects
joins
klu
pweek1
sys

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

```

43 (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
44 (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
45 (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
46
47 • SELECT * FROM Loans;
48
49 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
50 FROM Loans l
51 INNER JOIN Members m ON l.member_id = m.member_id
52 INNER JOIN Books b ON l.book_id = b.book_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Member_Name	Book_Title
John Smith	The Great Gatsby
John Smith	To Kill a Mockingbird
John Smith	1984
John Smith	The Great Gatsby
Emma Wilson	To Kill a Mockingbird
Emma Wilson	The Great Gatsby

Result 4 x | Read Only | Context Help | Snippets

Output: Action Output

#	Time	Action	Message	Duration / Fetch
✓ 6	15:47:19	CREATE TABLE Members (member_id INT PRIMAR...	0 row(s) affected	0.047 sec
✓ 7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
✓ 8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
✓ 10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
✓ 11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec

Step 12 — GROUP BY Query

Navigator: SCHEMAS
Filter objects
joins
klu
pweek1
sys

SQL File 3* SQL File 3* SQL File 4* SQL File 5* x

Limit to 1000 rows

```

48
49 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
50 FROM Loans l
51 INNER JOIN Members m ON l.member_id = m.member_id
52 INNER JOIN Books b ON l.book_id = b.book_id;
53
54 • SELECT published_year, COUNT(book_id) AS Total_Books
55 FROM Books
56 GROUP BY published_year
57 ORDER BY published_year;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

published_year	Total_Books
1925	1
1949	1
1960	1

Result 5 x | Read Only | Context Help | Snippets

Output: Action Output

#	Time	Action	Message	Duration / Fetch
✓ 7	15:47:47	INSERT INTO Members (member_id, full_name, email...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.047 sec
✓ 8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
✓ 10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
✓ 11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec

Step 13–16 — Donation via TRANSACTION (Atomicity)

Navigator

SQL File 3* SQL File 3* SQL File 4* SQL File 5*

Limit to 1000 rows

My Snippets

SCHEMAS

Filter objects

- joins
- klu
- pweek1
- sys

Administration Schemas

Information

No object selected

Context Help Snippets

```

48
49 • SELECT m.full_name AS Member_Name, b.title AS Book_Title
50 FROM Loans l
51 INNER JOIN Members m ON l.member_id = m.member_id
52 INNER JOIN Books b ON l.book_id = b.book_id;
53
54 • SELECT published_year, COUNT(book_id) AS Total_Books
55 FROM Books
56 GROUP BY published_year
57 ORDER BY published_year;
58
59 • CREATE TABLE Donation_History (
60     donation_id INT PRIMARY KEY,
61     book_id INT,
62     donor_name VARCHAR(100),
63     donation_date DATE,
64     FOREIGN KEY (book_id) REFERENCES Books(book_id)
65 );

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 8	15:48:03	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
✓ 10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
✓ 11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec

Object Info Session

SQL File 3*

SQL File 3*

SQL File 4*

SQL File 5* x

Limit to 1000 rows

My Snippets

SCHEMAS

Filter objects

joins

klu

pweek1

sys

Administration

Schemas

Information

No object selected

52

INNER JOIN Books b ON l.book_id = b.book_id;

53

54

SELECT published_year, COUNT(book_id) AS Total_Books

55

FROM Books

56

GROUP BY published_year

57

ORDER BY published_year;

58

59

CREATE TABLE Donation_History (

60

donation_id INT PRIMARY KEY,

61

book_id INT,

62

donor_name VARCHAR(100),

63

donation_date DATE,

64

FOREIGN KEY (book_id) REFERENCES Books(book_id)

65

);

66

67

START TRANSACTION;

68

69

INSERT INTO Books (book_id, title, isbn, published_year)

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	15:49:58	CREATE TABLE Loans (loan_id INT PRIMARY KEY...	0 row(s) affected	0.094 sec
10	16:22:00	INSERT INTO Loans (loan_id, member_id, book_id, l...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.000 sec
11	16:22:24	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec
14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec
15	16:25:33	START TRANSACTION	0 row(s) affected	0.000 sec

SQL File 3*

SQL File 3*

SQL File 4*

SQL File 5* x

Limit to 1000 rows

My Snippets

SCHEMAS

Filter objects

joins

klu

pweek1

sys

Administration

Schemas

Information

No object selected

58

59

CREATE TABLE Donation_History (

60

donation_id INT PRIMARY KEY,

61

book_id INT,

62

donor_name VARCHAR(100),

63

donation_date DATE,

64

FOREIGN KEY (book_id) REFERENCES Books(book_id)

65

);

66

67

START TRANSACTION;

68

69

INSERT INTO Books (book_id, title, isbn, published_year)

70

VALUES (4, 'Animal Farm', '9780451526342', 1945);

71

72

INSERT INTO Donation_History (donation_id, book_id, donor_name, donat

73

VALUES (1, 4, 'Raj Kumar', CURDATE());

74

75

COMMIT;

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	16:23:07	SELECT m.full_name AS Member_Name, b.title AS B...	10 row(s) returned	0.000 sec / 0.000 sec
13	16:23:50	SELECT published_year, COUNT(book_id) AS Total_...	3 row(s) returned	0.000 sec / 0.000 sec
14	16:24:55	CREATE TABLE Donation_History (donation_id l...	0 row(s) affected	0.047 sec
15	16:25:33	START TRANSACTION	0 row(s) affected	0.000 sec
16	16:25:53	INSERT INTO Books (book_id, title, isbn, published_y...	1 row(s) affected	0.000 sec
17	16:25:59	INSERT INTO Donation_History (donation_id, book_id, ...	1 row(s) affected	0.000 sec
18	16:26:02	COMMIT	0 row(s) affected	0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

Navigation

SCHEMAS

Filter objects

joins

klu

pweek1

sys

Administration

Schemas

Information

No object selected

Object Info

Session

SQL File 3*

SQL File 3*

SQL File 4*

SQL File 5* x

Limit to 1000 rows

My Snippets

70

VALUES (4, 'Animal Farm', '9780451526342', 1945);

71

72 •

INSERT INTO Donation_History (donation_id, book_id, donor_name, donat

73

VALUES (1, 4, 'Raj Kumar', CURDATE());

74

75 •

COMMIT;

76

77 •

CREATE INDEX idx_books_isbn ON Books(isbn);

78

79 •

SELECT * FROM Books WHERE isbn = '9780451524935';

Result Grid

Filter Rows:

Edit:

Export/Import:

Result Grid

Form Editor

	book_id	title	isbn	published_year
▶	3	1984	9780451524935	1949
*	NULL	NULL	NULL	NULL

Books 6 x

Apply

Revert

Context Help

Snippets

Output

Action Output

	#	Time	Action	Message	Duration / Fetch
✓	14	16:24:55	CREATE TABLE Donation_History (donation_id I...	0 row(s) affected	0.047 sec
✓	15	16:25:33	START TRANSACTION	0 row(s) affected	0.000 sec
✓	16	16:25:53	INSERT INTO Books (book_id, title, isbn, published_y...	1 row(s) affected	0.000 sec
✓	17	16:25:59	INSERT INTO Donation_History (donation_id, book_i...	1 row(s) affected	0.000 sec
✓	18	16:26:02	COMMIT	0 row(s) affected	0.000 sec
✓	19	16:26:43	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.078 sec
✓	20	16:26:51	SELECT * FROM Books WHERE isbn = '9780451524...	1 row(s) returned	0.000 sec / 0.000 sec